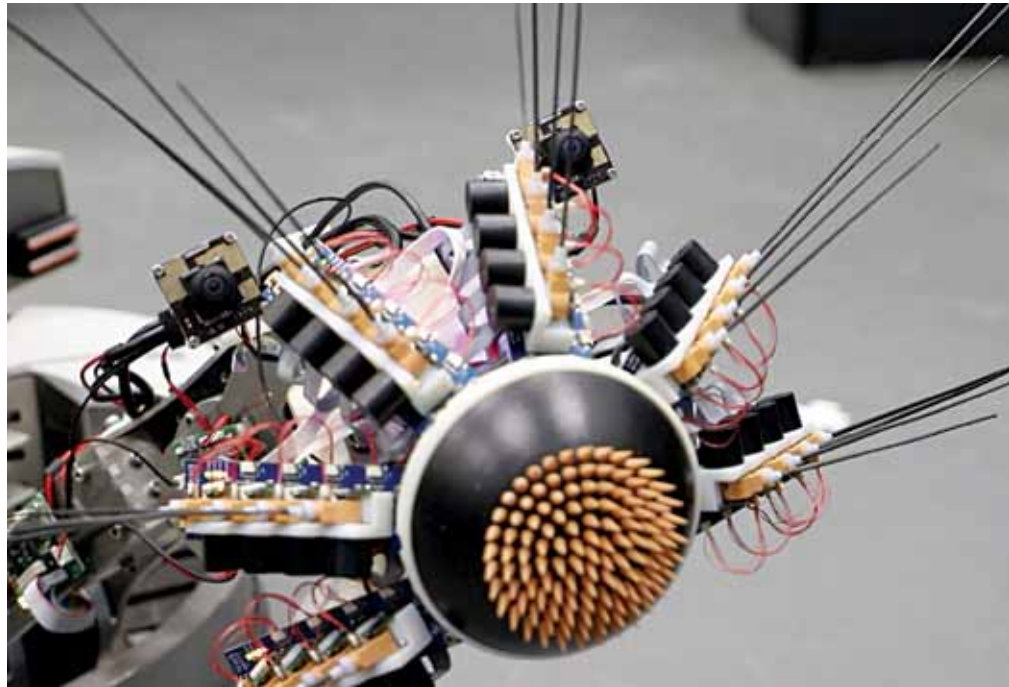




Robot combines touch and vision to navigate

In order to understand the neural mechanisms behind the brain's perception and navigation of the surrounding world, a team of cognitive neuroscientists, computational modellers, and roboticists from the EBRAINS Research Infrastructure have developed a robot called Whiskey. The rodent-like robot autonomously explores its environment with the help of head-mounted cameras that serve as eyes and 24 artificial whiskers that gather tactile information. By leveraging simultaneous localisation and mapping, the robot maps an object or environment using touch, converting any relative self-motion into an internal representation. The brain based model operating the robot has an edge over traditional deep learning systems, especially when it comes to navigation and recognition of familiar scenes.



Brain models are helping to discover better robot navigation methods (Credit: <https://ebrains.eu>)



Plant patch monitors crop diseases



Plant patch enables continuous monitoring for crop diseases (Credit: <https://news.ncsu.edu>)

Researchers from North Carolina State University have developed a patch that plants can 'wear' to continuously monitor plant diseases or other stresses such as crop damage or extreme heat. By measuring the volatile organic compounds (VOCs) emitted by plants under different circumstances, the sensors in the patch can alert users to specific problems. The sensors are coated with various chemical ligands that respond to the presence of specific VOCs, allowing the system to detect and measure VOCs in gases emitted by the plant's leaves. The rectangular patches are 30mm long and have a flexible material containing graphene based sensors and flexible silver nanowires.